DGMS Examination 2019 — Question 52 Detailed Solution

Junior Scientific Officer (Geology) | Hydrogeology & Hillslope Hydrology

QUESTION 52

The lateral unsaturated flow of water in the soil zone, where a highly permeable geological unit overlays a less permeable geological unit, is termed as:

(a) Overland flow	(b) Saturated overland flow
(c) Sheet flow	**(d) Throughflow** CORRECT ANSWER

1. Executive Summary & Answer Verification

The correct option is **(d) Throughflow** (also referred to in hydrological literature as *interflow* or *subsurface stormflow*).

In catchment hydrology and hydrogeology:

- When precipitation infiltrates into the ground, water percolates vertically through the porous, highly permeable upper soil zone (A-horizon / vadose zone).
- When this infiltrating water encounters a lower-permeability boundary (such as an argillic B-horizon, fragipan, glacial till, or unweathered bedrock where hydraulic conductivity drops sharply: $K_{\text{upper}} \gg K_{\text{lower}}$), downward drainage is impeded.
- A perched hydraulic gradient is established parallel to the slope. Water is deflected laterally, moving downslope through the unsaturated soil matrix and root macropores toward the channel network. This lateral soil-water flux is universally termed **Throughflow**.

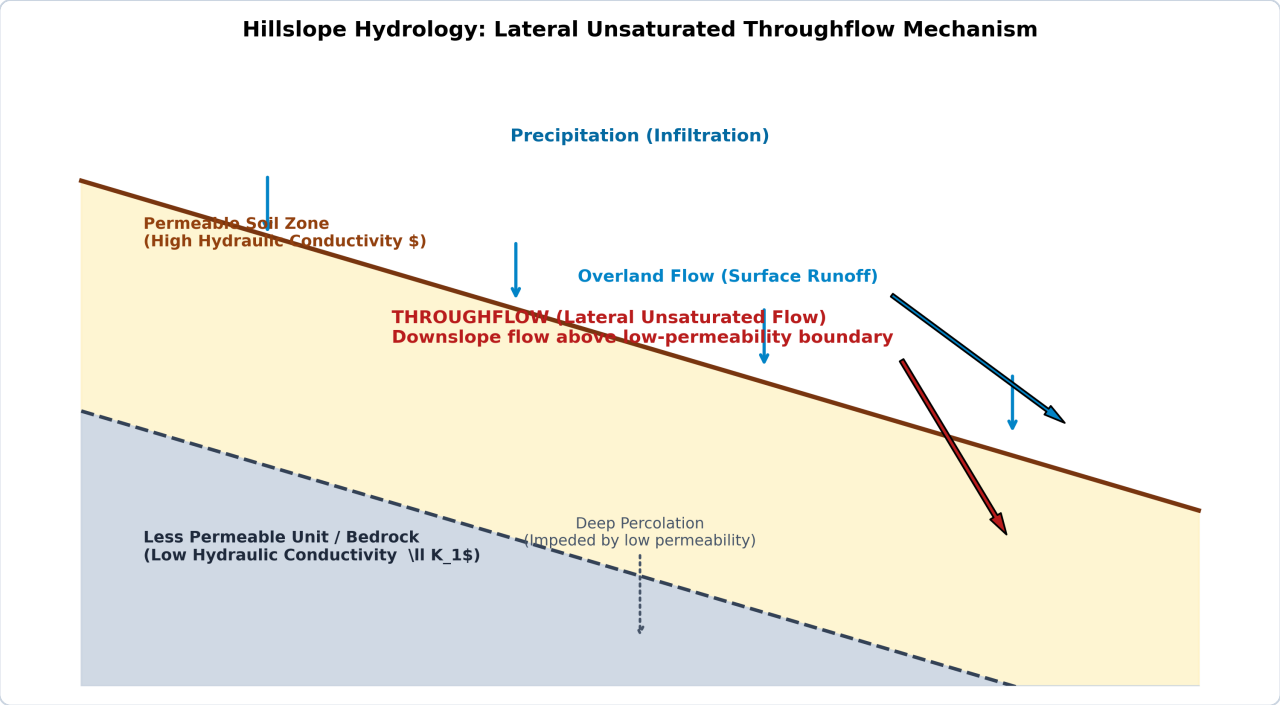


Figure 52.1: *Hillslope hydrological pathways illustrating* ***Throughflow****: Infiltrated water encounters a permeability barrier ($K_1 \gg K_2$) and is deflected laterally downslope through the soil zone, discharging into stream channels without initially traveling as surface runoff.*

2. Comprehensive Hydrological Pathway Analysis

2.1 Mechanics of Throughflow (Interflow)

Throughflow was first thoroughly quantified by M. J. Kirkby and R. J. Chorley (1967). Its physical behavior is described by Darcy's Law adapted for unsaturated/partially saturated porous media:

```
q = -K(\theta) \cdot \frac{\partial h}{\partial s}
```

Where:

- q is the lateral flux per unit cross-sectional area;
- $K(\theta)$ is the unsaturated hydraulic conductivity, which increases exponentially as soil volumetric moisture content (θ) rises;
- $\frac{\partial h}{\partial s}$ is the hydraulic head gradient along the sloping interface.

Because the underlying stratum has very low conductivity ($K_{\text{lower}} \approx 0$), water accumulates above the interface, producing a transient saturated wedge that accelerates lateral throughflow through soil pipes, burrows, and root channels.

2.2 Evaluation of the Other Flow Pathways

1. **(a) Overland Flow (Hortonian Overland Flow):** Occurs directly on the ground surface when precipitation intensity exceeds the infiltration capacity of the soil ($P > f$). It does *not* involve flow within the soil zone.
2. **(b) Saturated Overland Flow (Dunne Flow):** Occurs when the water table rises to intersect the ground surface (e.g., at the toe of a slope or in valley bottoms), saturating the entire soil column (100% saturation) so that any additional rainfall cannot infiltrate and runs off over the surface.

3. **(c) Sheet Flow:** Refers to an unconfined, continuous thin sheet of water moving across an unchannelized surface (such as a pediment or asphalt surface). It is a surface runoff mechanism, not subsurface soil flow.

3. Comparison Table of Runoff Generation Mechanisms

Pathway	Flow Location	Saturation State	Governing Physical Mechanism
Hortonian Overland Flow	Ground surface	Soil surface partially or fully wet	Rainfall rate exceeds soil infiltration capacity ($P > f$)
Saturated Overland Flow	Ground surface	100% fully saturated soil column	Water table intersects surface; zero infiltration capacity remaining
Sheet Flow	Ground surface	Unconfined surface sheet	Shallow overland flow before concentrating into rills
Throughflow (Interflow)	**Subsurface (Soil zone)**	**Unsaturated to locally perched**	**Lateral flow above lower-permeability boundary ($K_1 \gg K_2$)**
Baseflow	Deep phreatic zone	Fully saturated aquifer	Regional groundwater discharge sustaining perennial stream base

Textbook Authority

R. Allan Freeze & John A. Cherry, *Groundwater* (Prentice Hall):
"Throughflow (or interflow) is the lateral flow of water in the unsaturated zone that moves downslope through the soil zone to emerge into stream channels... It develops primarily where a permeable surficial soil horizon overlays a less permeable substrate."
M. J. Kirkby, *Hillslope Hydrology* (Wiley):
"Throughflow refers to the lateral unsaturated or perched saturated movement of water through the soil profile towards the stream, promoted by permeability contrasts between soil horizons."